



Disclaimer: This file is based completely on my opinions and experience. Everything you read here could be disagreed with. While I do recommend a team to know all of the systems written in this file, the order in which every system is written is merely a recommendation on how I would teach them. I highly encourage teams to search about more systems that are not specified in this file.

Useful FRC systems ranked by system type:

Intakes

- 111 - Pivoting intake
- 112 - Cut bumper intake
- 113 - Under The Bumper
- 121 - Static Claw
- 122 - Passive claw
- 123 - Active Claw
- 130 - Rack and Pinion
- 141 - 4 Bar Linkage
- 142 - 6 Bar Linkage

Indexers

- 211 - Spinning Indexer
- 212 - Dual Spinning Indexer
- 221 - Belt Indexer
- 222 - Roller Indexer
- 231 - Dye Rotor Indexer
- 232 - Tunnel Indexer

Outtake

- 311 - Static Drum/Line Shooter
- 312 - Hooded Shooter
- 313 - Adjustable Line Shooter
- 321 - Passive Dumper
- 322 - Active Dumper



Extension Mechanisms

411 - Continuous Elevator

412 - Cascading Elevator <3L

413 - Cascading Elevator >3L

421 - Telescopic Arm

End Effector Manipulation

510 - Turret

521- 3DOF Arm

522 - 4+DOF Arm

Misc

910 - Ratchet

921 - Differential Drive

922 - Differential Gearbox

-  - Easy to understand and easy to model
-  - Medium level, harder concept and a more complicated model
-  - Hard concept to grasp, difficult to model
-  - Only for the bestest teams